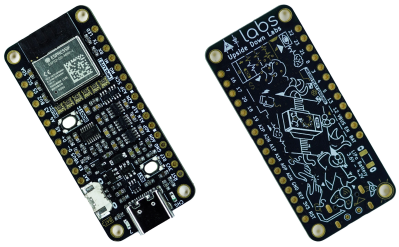
# Neuro PlayGround (NPG) Lite

An expandable 3-channel wireless BioAmp for HCI & BCI

## Overview

Neuro PlayGround Lite (NPG Lite) is a multichannel Adafruit feather form-factor wireless bio-physiological signal acquisition board. It can be used for Electrocardiography (ECG), Electromyography (EMG), Electrooculography (EOG), or Electroencephalography (EEG). Its compact footprint and hassle-free setup ensure portability, quick deployment and a clutter-free experience, making it ideal for research, education and wearable applications.



## Features

- Expandable with NPG Playmate Boards and 3rd party FeatherWing boards, adding features like a vibration motor for haptic feedback, a buzzer for audio feedback, and I2C connectivity
- Works with the open source, cross-platform Chords software suite (Chords-Web, Chords-Python, Chords LSL Connector and Chords LSL Visualizer) to visualize signals in real-time, apply filtering, perform FFT analysis, and much more
- Dedicated NPG Lite Flasher tool to flash pre-built firmware binaries from GitHub or your own binary files
- Bluetooth, Zigbee, Wi-Fi or Thread for mesh networking, to capture bio-physiological signals how you want and where you want it

## Specifications

| | |
|---|---|
| **Bio-signal Channels** | 3-6 Channel BioAmp (ECG, EMG, EOG, EEG) |
| **Microcontroller** | ESP32-C6 (RISC-V) with 4 MB Flash |
| **Biopotential(s)** | EXG |
| **ADC Resolution** | 12 Bit |
| **Wireless I/O** | Wi-Fi 6, Bluetooth LE 5, Thread, Zigbee |
| **USB & Programming** | USB-C for power, charging & programming; UART, JTAG |
| **LEDs** | 6 addressable RGB LEDs, User LED, Charging indicator LED |
| **Buttons** | Reset button + User/Boot Buttons |
| **Power Options** | USB-C or LiPo battery |
| **Power Regulation** | 600 mA LDO with LiPo battery charging onboard |
| **Battery Connector** | PicoBlade LiPo |
| **Form Factor** | Standard Adafruit Feather (GPIO, Power, RX/TX pins) |
| **Protection** | On-board patient & USB ESD protection |
| **Compatible Hardware** | NPG Playmates & FeatherWing boards & QWIIC |

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# 1 Neuro PlayGround (NPG) Lite

**An expandable 3-channel wireless BioAmp for HCI & BCI**

## 1.4 Board Layout & Pinout Diagram

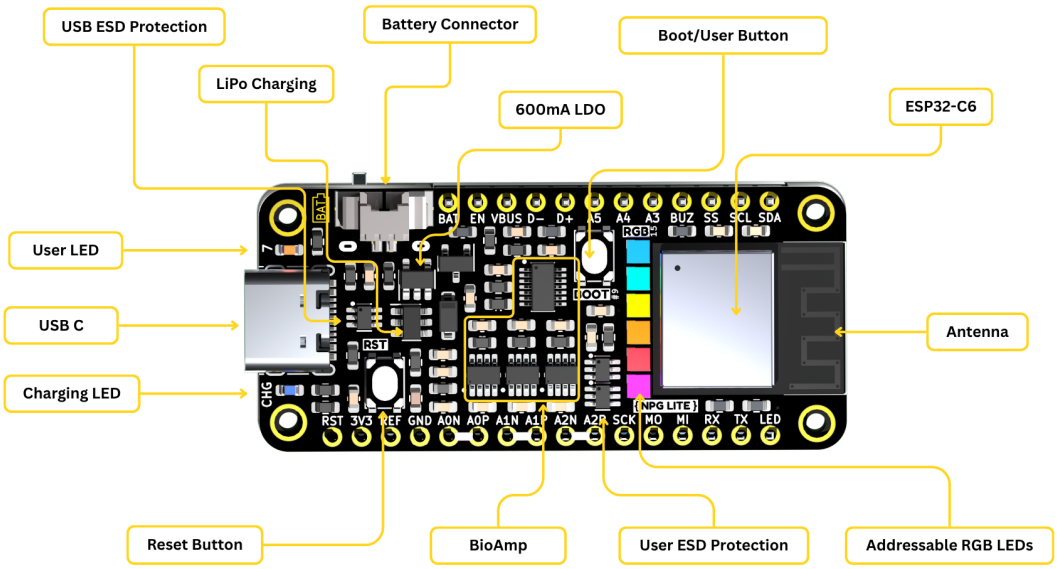



Fig. 1: Neuro PlayGround (NPG) Lite annotations

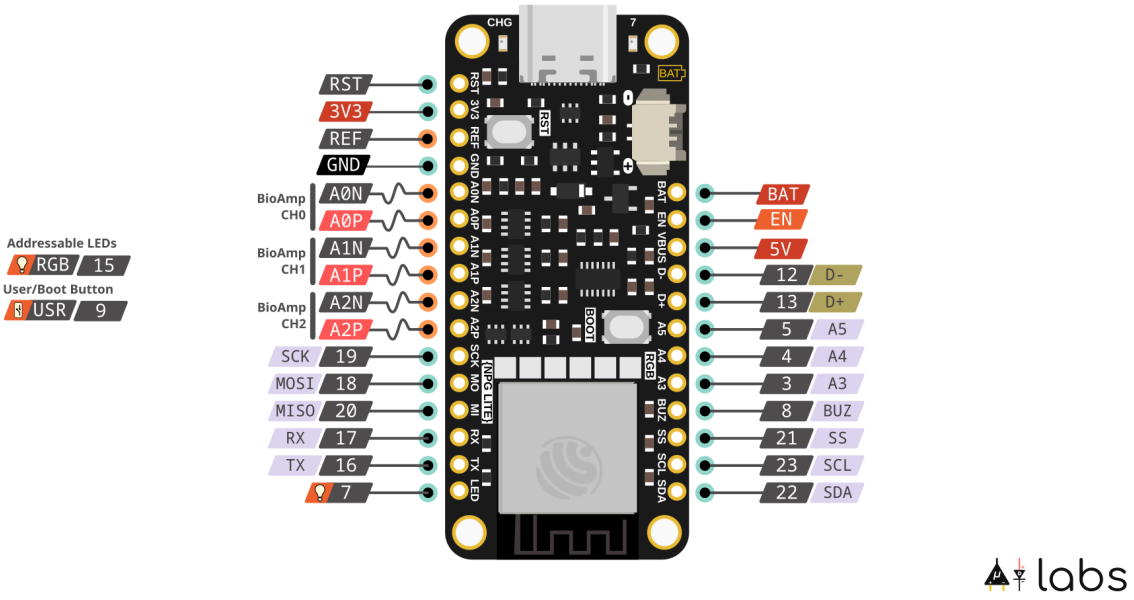



Fig. 2: Neuro PlayGround (NPG) Lite Pinout

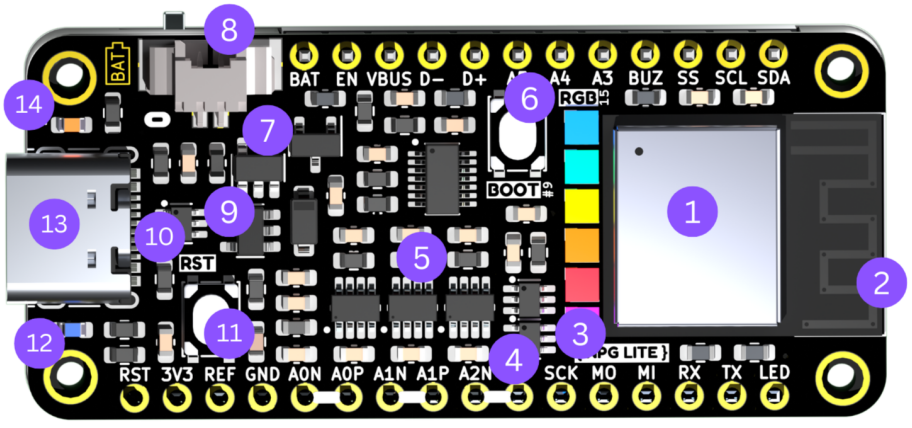



Fig. 3: Neuro PlayGround (NPG) Lite Components

## 1.5 Components

| Number | Components | Description |
|---|---|---|
| 1 | **ESP32-C6** | The ESP32-C6 is a Wi-Fi 6 and Bluetooth 5 (LE) enabled microcontroller developed by Espressif Systems. |
| 2 | **Antenna** | The onboard PCB antenna is tuned for 2.4GHz, ensuring strong, stable Wi-Fi and Bluetooth connectivity with low power use. |
| 3 | **6 Addressable RGB LEDs** | Enables visualization of device states, signal strength, or neural feedback using dynamic color indicators. |
| 4 | **User ESD Protection** | ESD protection components, ensuring durability and reliability when connecting electrodes. |
| 5 | **BioAmp** | The onboard BioAmp is a low-noise, high-gain biopotential amplifier that captures and amplifies EMG, ECG, and EEG signals for digital processing by the ESP32-C6. |
| 6 | **Boot/User Button** | The Boot/User button enables bootloader mode for firmware flashing and serves as a programmable user input. |
| 7 | **600mA LDO** | The voltage regulator converts higher input voltages (USB or battery) into a stable 3.3V output. |
| 8 | **Battery Connector** | It offers a safe, port for battery power, allowing portable operation of the board. **Note:** There is no polarity protection. |
| 9 | **LiPo Charging** | The integrated LiPo charging controller safely manages charging cycles, supporting automatic termination and trickle charging for optimal battery health. |
| 10 | **USB ESD Protection** | USB ESD protection shields the interface from electrostatic and voltage spikes, ensuring reliable data transfer and preventing component damage. |
| 11 | **Reset Button** | The reset button restarts the microcontroller and running program without power loss, aiding debugging and firmware testing. |
| 12 | **Charging LED** | The Charging LED indicates battery charging status, lighting during charging and turning off when fully charged for immediate feedback. |
| 13 | **USB C** | USB C port for data transfer and power input. |
| 14 | **User LED** | A programmable LED on a GPIO pin used to show board activity, connection status, or custom signals during development. |

## 1.6 Explorer, Ninja, and Beast!

If you're comfortable with soldering and only need a 3-channel wireless BioAmp, the **Explorer Kit** is a great choice. For the most complete NPG Lite experience right out of the box, go with **Ninja Pack**. If you require additional channels for full-body signal acquisition or advanced experimentation, the **Beast Pack** offers the most comprehensive configuration.

| Pack Options | Explorer | Ninja | Beast |
|---|---|---|---|
| NPG Lite | Included | Included | Included |
| Playmate Type | Proto | VibZ | VibZ+ |
| Channel Count | 3 Channels | 3 Channels | 6 Channels |
| Gel Electrodes | 12 Pieces | 24 Pieces | 24 Pieces |
| Snap Cables | 7 Pieces | 7 Pieces | 13 Pieces |
| Alligator Cables | 5 Pieces | 5 Pieces | 8 Pieces |
| Battery Cable | Included | Included | Included |
| LiPo Battery | Not Included | Included | Included |
| 3D Printed Case | Not Included | Included | Included |
| USB Cable | Not Included | Included | Included |

## 1.7 Playmates

Playmates are modular expansion boards designed to enhance the functionality of Neuro PlayGround (NPG) Lite. These add-ons seamlessly integrate with the core system, allowing users to extend their capabilities to create awesome Human-Computer Interface (HCI) & Brain-Computer Interface (BCI).

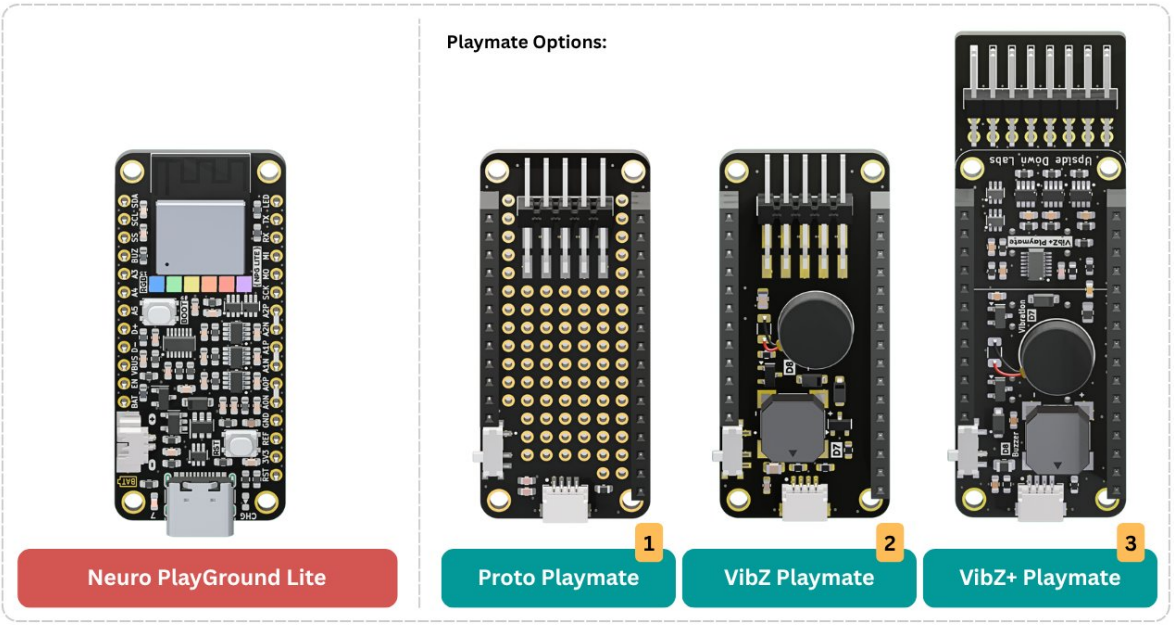



Fig. 4: Neuro PlayGround (NPG) Lite Playmates

### 1.7.1 Need of Playmate

Playmate offer a convenient way to expand the functionality of your NPG Lite setup without complex wiring or modifications. It easily plug into NPG Lite's Feather compatible headers, providing additional features such as

#### 1. Proto Playmate

a. ON/OFF switch for power control
b. QWIIC/Stemma QT port
c. Electrodes connector
d. Large prototyping area for custom circuits

#### 2. VibZ Playmate

a. ON/OFF switch for power control
b. QWIIC/Stemma QT port
c. Electrodes connector
d. Vibration motor for haptic feedback
e. Buzzer for audio feedback

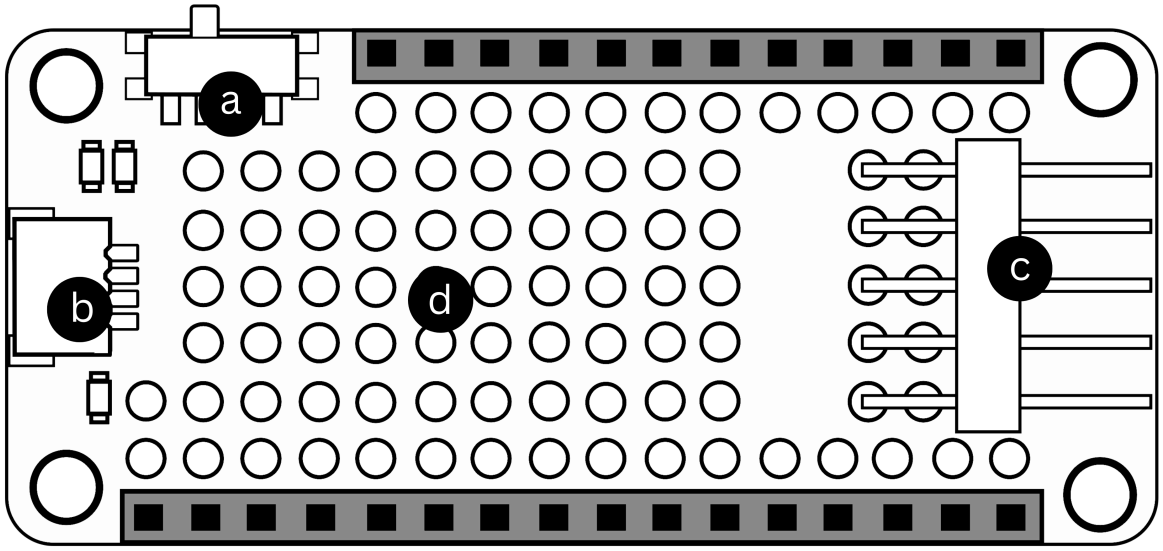



Fig. 5: Neuro PlayGround (NPG) Lite Proto Playmate

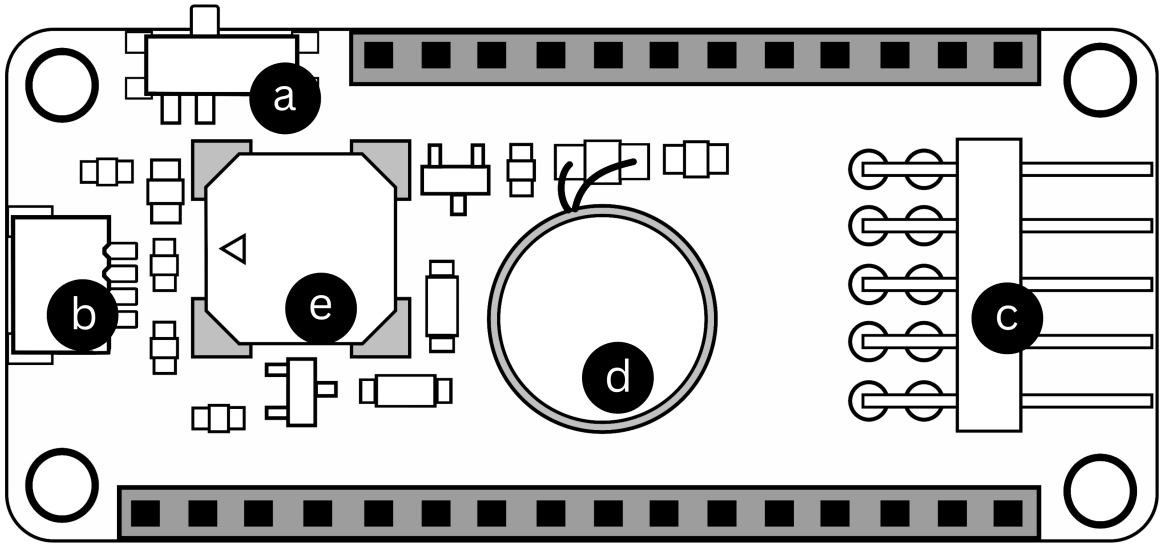



Fig. 6: Neuro PlayGround (NPG) Lite VibZ Playmate

### 3. VibZ+ Playmate

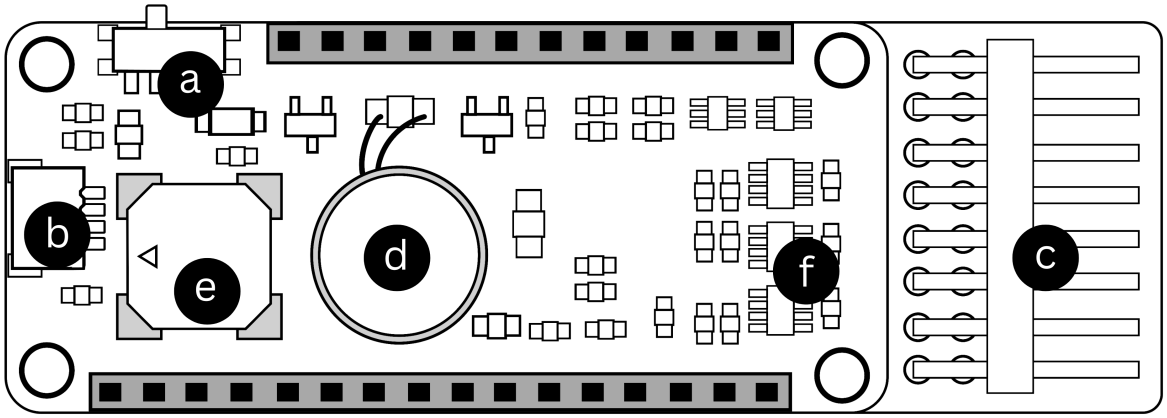



Fig. 7: Neuro PlayGround (NPG) Lite VibZ+ Playmate

a. ON/OFF switch for power control
b. QWIIC/Stemma QT port
c. Electrodes connector
d. Vibration motor for haptic feedback
e. Buzzer for audio feedback
f. Additional 3 channel BioAmp (total 6 channels)

## 1.7.2 Comparison of Playmates

| Feature | Proto | VibZ | VibZ+ |
|---|---|---|---|
| **Bio Channels** | 3 | 3 | 6 |
| **Prototyping Area** | Yes(Large) | No | No |
| **Haptic Feedback** | No | Yes (Vibration) | Yes (Vibration) |
| **Audio Feedback** | No | Yes (Buzzer) | Yes (Buzzer) |
| **QWIIC Port** | Yes | Yes | Yes |
| **Power Switch** | Yes | Yes | Yes |
| **Electrode Headers** | Dupont 2.54 mm | Dupont 2.54 mm | Dupont 2.54 mm |
| **Common Negative** | Solder option | Solder option | Switch |
| **Best For** | Development | Complete Projects | Research |

## 1.8 Usage Guide

To help you get started quickly, we've prepared a step-by-step video tutorial. This guide walks you through setup, connections, and running your experiments with the Neuro PlayGround Lite.

https://youtu.be/hBrbw9vF-m4

**Troubleshooting**

Having trouble connecting your **NPG Device** with Chords Web?

If you try to connect the device and the pop-up window for **Bluetooth Connection** doesn't appear, make sure your chromium based browser has following feature Enabled:

1. Open a new tab and go to `chrome://flags`
2. Find and enable the following flags:
   - **Web Bluetooth**
   - **Use the new permissions backend for Web Bluetooth**
   - **Web Bluetooth confirm pairing support**
   - **Experimental Web Platform features**

## 1.9 Compatible Software

### 1.9.1 Flasher Tools

#### NPG Lite Flasher Web

- NPG Lite Flasher Web is a browser-based flashing tool for the Neuro PlayGround Lite and other ESP32-based boards. It needs no Arduino IDE or additional setup: connect your device, select the BLE, Wi-Fi or Serial firmware you need, and flash it directly from the browser in seconds.
- To know more about NPG Lite Flasher Web watch the YouTube video.

https://youtu.be/sKYY074Yg3A

#### NPG Lite Flasher App

- NPG Lite Flasher is a cross-platform, flashing utility for the Neuro PlayGround Lite board, providing seamless firmware updates over serial or DFU via a simple CLI.
- To know more about NPG Lite Flasher watch the YouTube video.

https://youtu.be/NDk-P3nob_0

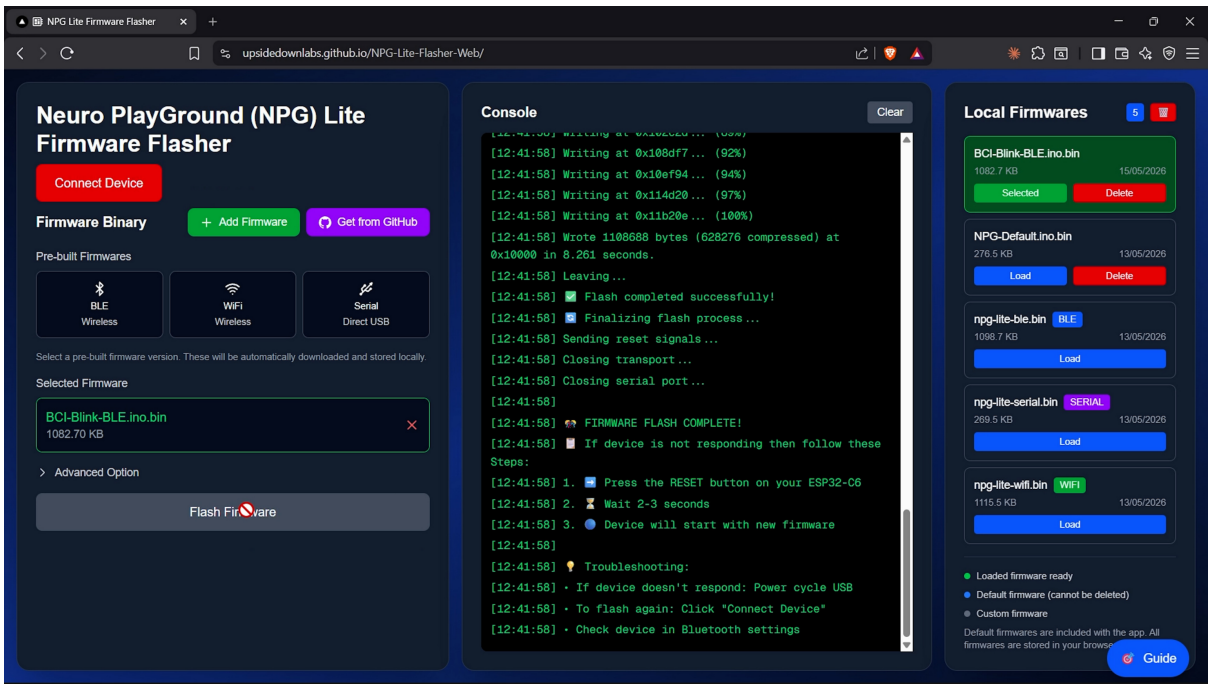



Fig. 8: NPG Lite Flasher Web Landing Page

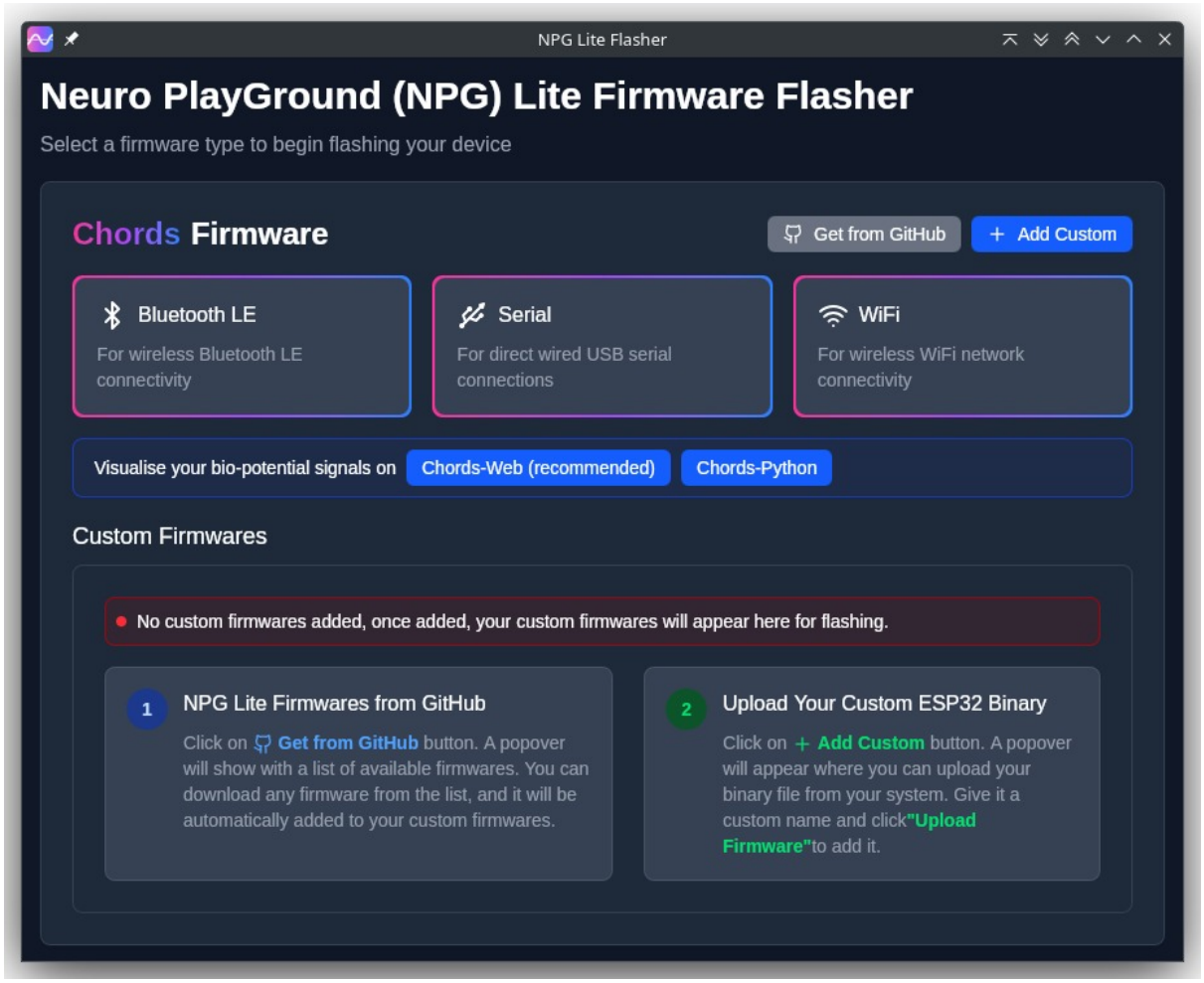



Fig. 9: NPG Lite Flasher Landing Page

### 1.9.2 Chords Web

- Visit Chords Web for real-time biopotential signal visualization (EEG, EMG, ECG, EOG), featuring advanced plotting, filtering, snapshotting and CSV recording. Flash the Chords-Arduino-Firmware on your board to visualize signals on Chords Web.

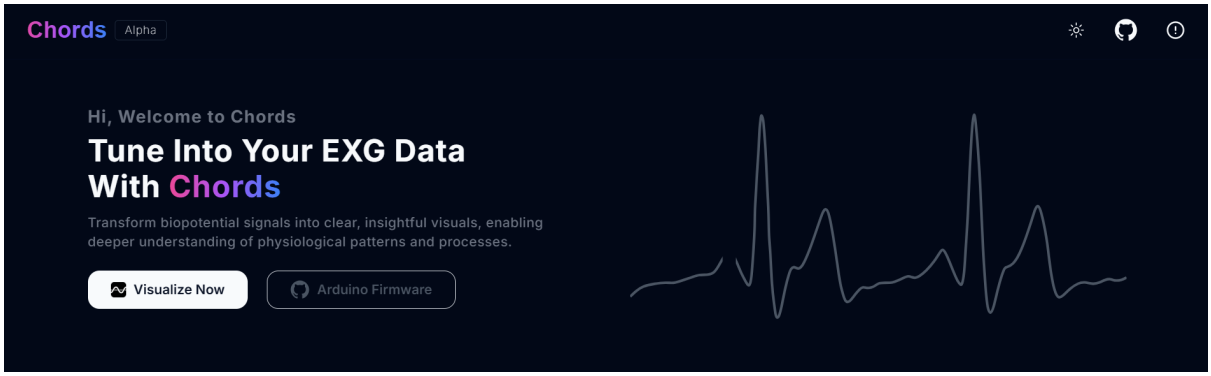



Fig. 10: Chords Web Landing Page

- To know more about Chords Web click here or watch the YouTube video.

https://youtu.be/Tqor4mo2Ji8

### 1.9.3 Chords Python

- Chords Python is an open-source bag of tools for recording biopotential signals like ECG, EMG, EEG, or EOG, along with visualization using BioAmp hardware.

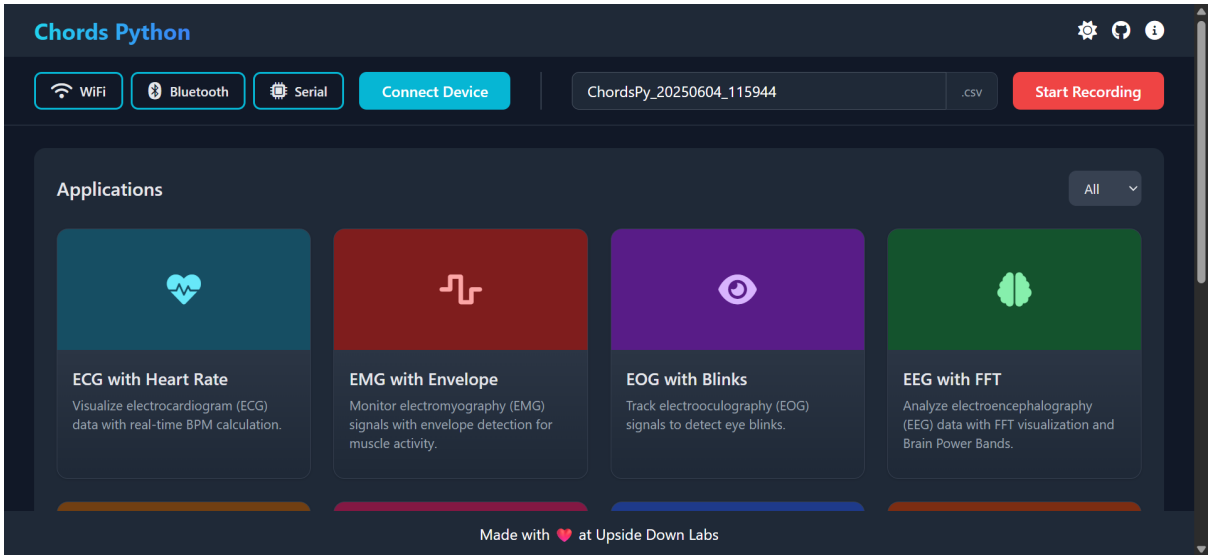



Fig. 11: Chords Python Landing Page

- To know more about Chords Python watch the YouTube video.

https://youtu.be/TiDwSQEY2eY

### 1.9.4 Chords LSL Connector

- Chords LSL Connector is a Rust based bridge that streams data from devices running Chords firmware into the Lab Streaming Layer (LSL), enabling synchronized acquisition and analysis with BCI/EEG software.

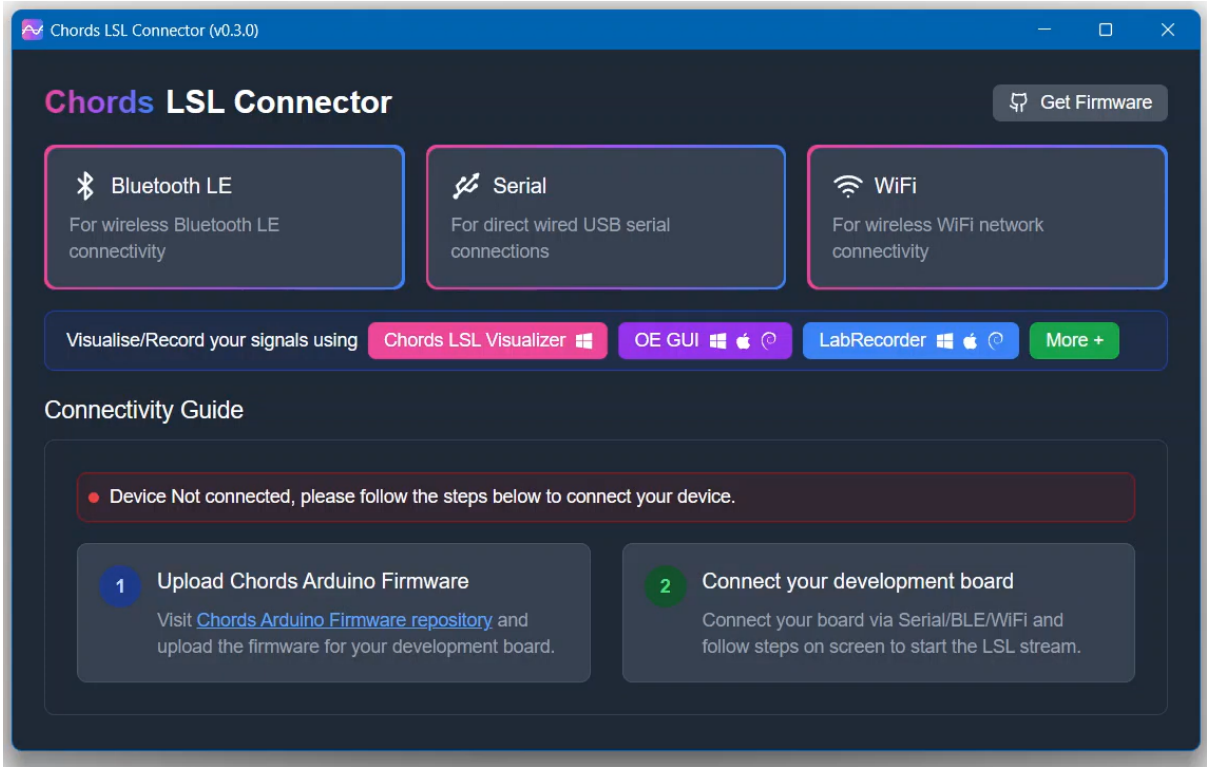



Fig. 12: Chords LSL Connector Landing Page

- To know more about Chords LSL Connector watch the YouTube video.

https://youtu.be/94lPImB0eRo

### 1.9.5 Chords LSL Visualizer

- Chords LSL Visualizer is a Rust based program to visualize biopotenial signals from any LSL stream.

## 1.10 Project Tutorials

NPG Lite, with its multi-channel BioAmp, wireless connectivity, battery operation, and expansion interface, allows users to create HCI and BCI applications with ease. We've created projects leveraging ECG, EMG, EOG, and EEG signals to inspire you, but the possibilities cover so many more use cases.

**Note**

Each of these projects runs on the `BLE firmware`, which you can flash using the following flasher tools:

- For flashing BLE Firmware, you may use either the NPG Lite Flasher Web or the NPG Lite Flasher App.
- Alternatively, you can use Arduino IDE to manually upload the firmware.
- It is recommended to use ESP32 version `3.2.0` to ensure error free operations.

Fig. 13: Chords LSL Visualizer Landing Page

1. Make Muscular Music  This is a human-computer interface (HCI) that uses 3-channel EMG data to detect muscle signals from the left hand, right hand, and chest to control virtual musical instruments. When muscles contract, each channel triggers a distinct sound effect, transforming your body into an organic orchestra.

https://youtu.be/O36z_YhaDqg

**Note**

- To learn about this project, visit our Instructables page for detailed guide: Muscle Melody: Create Music With Your Muscle Movements (EMG)
- Visit the Web Interface to visualize in real time.

2. Track Your Ticker  This project focuses on capturing electrocardiogram (ECG) signals to calculate heart rate using Neuro PlayGround Lite and displaying them in a Chrome-based browser on a mobile phone in real-time via a Bluetooth LE connection.

https://youtu.be/2UDBqEsjJSI

**Note**

- To learn about this project, visit our Instructables page for detailed guide: Monitor ECG and Heartrate on Your Mobile Phone
- Visit the Web Interface to visualize in real time.

3. Jump In the Blink of an Eye  Instead of using the keyboard, you can control a game simply by blinking. The setup detects EOG signals from the eyes, sends the data over Bluetooth LE to the PC, detects eye blinks, and then takes the eye blink as a trigger to emulate the spacebar keystroke. You can configure the code to simulate any other keystroke as well.

https://youtu.be/ZJmUUtHJj08

**Note**

- To learn about this project, visit our Instructables page for detailed guide: Controlling Keyboard With Eye Blinks Using Neuro PlayGround Lite

4. Pop Bubbles With Your Mind This project showcases a brain-computer interface (BCI) application that utilizes EEG (electroencephalography) signals to operate an interactive bubble-popping game in a web browser. The system identifies beta wave activity (12–30 Hz), indicating concentration and attention levels. Bubbles appear and pop as the user maintains focus, but they cease to pop when concentration wanes.

https://youtu.be/MGTNjFtikZE

**Note**

- To learn about this project, visit our Instructables page for detailed guide: Pop Bubbles With Your Mind (EEG) | Neuro PlayGround (NPG) Lite
- Visit the Web Interface to visualize in real time.

5. CortEX CortEX is an open-source neurofeedback meditation platform that combines EEG and ECG signals to provide real-time feedback on your mental and emotional state. Designed to enhance mindfulness, it visualizes brainwave activity, heart rate, and hemisphere balance to help you meditate more effectively - all from your browser, using the NPG Lite device.

https://youtu.be/O3F-0UWxNfA

**Note**

- To learn about this project, visit our Instructables page for detailed guide: A Neurofeedback Meditation Journey
- Visit the Web Interface to visualize in real time.

6. Brain-Controlled Car: EEG + EMG Robotic Control The Brain-Controlled Car transforms brainwaves (EEG) and muscle activity (EMG) into real-time robotic control, letting you steer a car without traditional remotes - just focus your mind to move forward, flex your left or right arm to turn, or both arms to reverse. Powered by the Neuro PlayGround Lite, a wireless biopotential device, the system captures your signals via electrodes and transmits them over BLE to an ESP32-based receiver that drives the motors. This project offers a hands-on glimpse into the future of human–machine interaction, where thoughts and movements seamlessly command machines.

https://youtu.be/US_cDCRjV6c

**Note**

- To learn about this project, visit our Instructables page for detailed guide: Brain-Controlled Car :EEG + EMG Robotic Control

7. Playing GTA V with Your Focus & Muscles Control GTA V using your mind! The Brain-Controlled Gaming system uses the wireless Neuro PlayGround Lite to translate your brainwaves (EEG) and muscle contractions (EMG) into game controls. Focus to accelerate, and flex your left or right arm to steer. This BCI project delivers a mind-bending glimpse into the future of gaming, letting you command your car with nothing but your body signals.

https://youtu.be/LhvBo8MP7iQ

8. Cuboid Cuboid is a neurofeedback-powered game that turns your focus into action. Using real-time brainwave data from the NPG Lite device, the game challenges you to move a geometric cuboid upward by maintaining mental focus. With multiple difficulty levels, live EEG streaming, and visual feedback, Cuboid transforms concentration into an immersive, brain-controlled experience.

**Note**

- Visit the Web Interface to visualize in real time.

9. Muscle Strength  Muscle Strength Game transforms your physical efforts into insightful data. Simply connect to any muscle group you want to focus on, and you'll instantly visualize the raw electrical activity your muscles generate. But it's more than just raw data; the game also processes this information to display your real-time muscle strength. This allows you to track your progress, identify areas for improvement, and engage with your workouts in a whole new, data-driven way. Whether you're an athlete, rehabilitating an injury, or just curious about your body, this project offers a unique window into your muscular capabilities.

**Note**

- Visit the Web Interface to visualize in real time.

10. Mind-to-Words Using EEG  Mind-to-Words (M2W) application is an innovative Assistive Communication tool that enables users to control a menu interface solely through eye blinks. It utilizes the EOG artifacts present in a single channel EEG signal, which is captured by the NPG Lite device. The core functionality relies on a novel algorithm that separates and recognizes distinct control signals: a double blink is used to activate or switch between options, while a triple blink is used to confirm and make a selection.

https://youtu.be/UyXsqCrVxh4

**Note**

- M2W Web Application: Web Interface

### Neuro-PlayGround Lite C3 (Obsolete)

### Overview

Neuro Playground (NPG) Lite is a multichannel wireless bio-potential signal amplification device designed for recording EMG, ECG, EOG, and EEG. It comes in a compact Adafruit feather form factor and offers WiFi/BLE wireless connectivity. With the addition of daughter boards called Playmates, users can easily enhance the device's capabilities, enabling the students/researchers/hobbyists to create awesome Human-Computer Interface (HCI) and Brain-Computer Interface (BCI) applications.

### Features & Specifications

| | |
|---|---|
| No. of channels | 3 |
| Power options | USB type C or LiPo battery |
| Microcontroller | ESP32-C3 (RISC-V) with 4MB Flash |
| Power supply | 600mA LDO with filtering |
| Connectivity options | WiFi, Bluetooth, USB |
| Debugging options | JTAG, UART |
| LEDs | User LED, charging LED, 6 addressable RGB LEDs |
| Buttons | Reset & user button |
| BioPotentials | EMG, ECG, EOG and EEG |
| Compatible hardware | NPG Playmates & FeatherWing boards |
| Protection | Built-in patient & USB ESD protection |

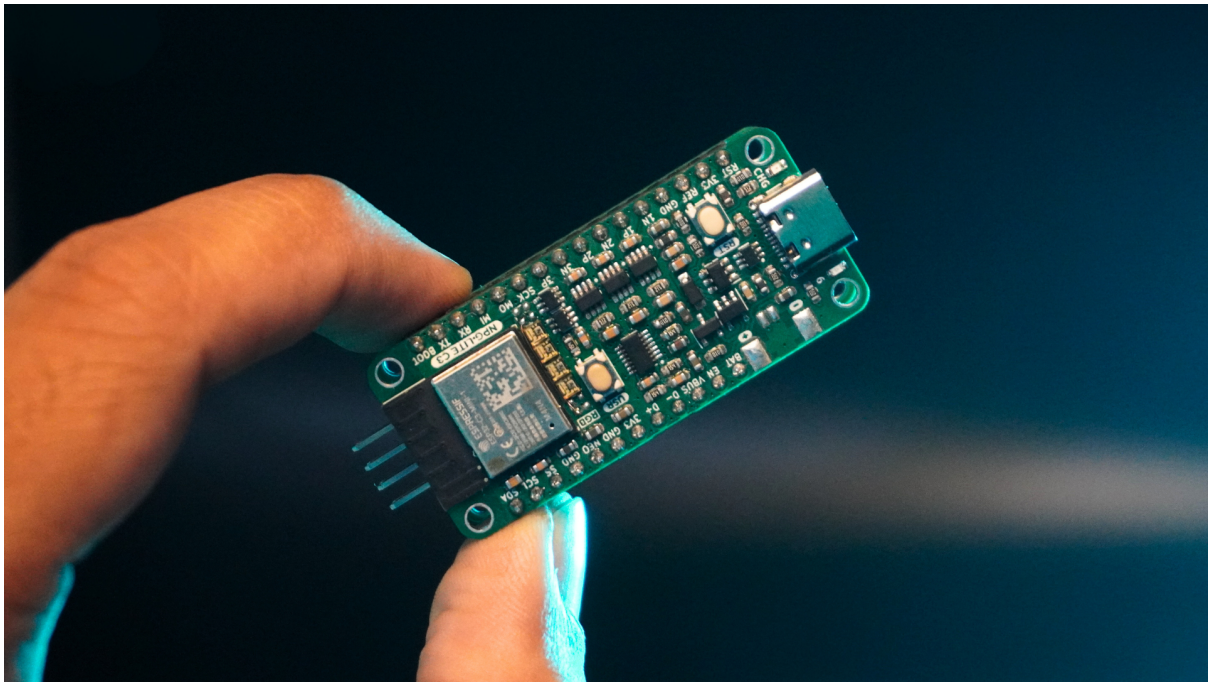


Fig. 14: Neuro PlayGround (NPG) Lite C3

## Board Layout

## Playmates

Playmates are modular expansion boards designed to enhance the functionality of **Neuro Play Ground (NPG) Lite**. These add-ons seamlessly integrate with the core system, allowing users to extend their capabilities to create awesome Human-Computer Interface (HCI) & Brain-Computer Interface (BCI).

### VibZ

The first Playmate, **VibZ**, introduces haptic feedback, a buzzer, a QWIIC port for easy sensor integration, an ON/OFF switch, and electrode header pins. With these Playmates, users can customize their setup for specific applications, whether in research, education, or interactive biofeedback experiments. This modular approach ensures flexibility, scalability, and better user experience.

## Features & Specifications

| |
|---|
| 2.54mm Dupont header pin for electrode connection |
| Vibration motor for haptic Feedback motor |
| Buzzer for audio feedback |
| QWIIC / STEMMA QT I2C port |
| ON/OFF Switch |

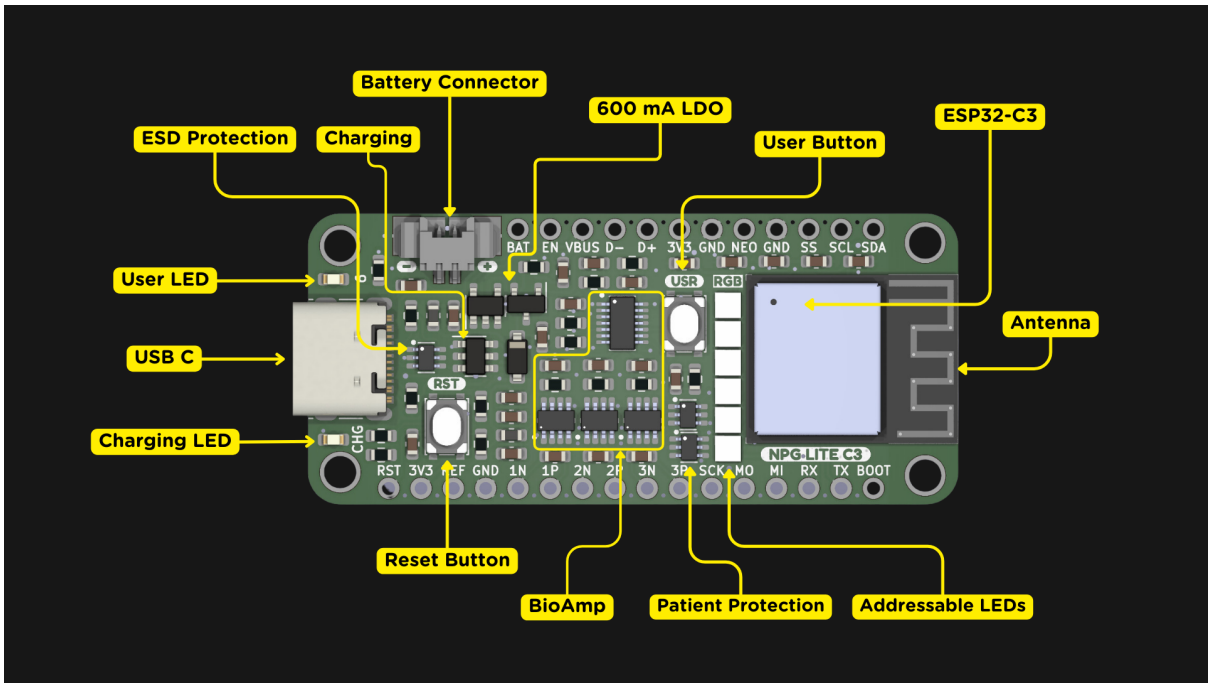



Fig. 15: Neuro PlayGround (NPG) Lite C3 annotations

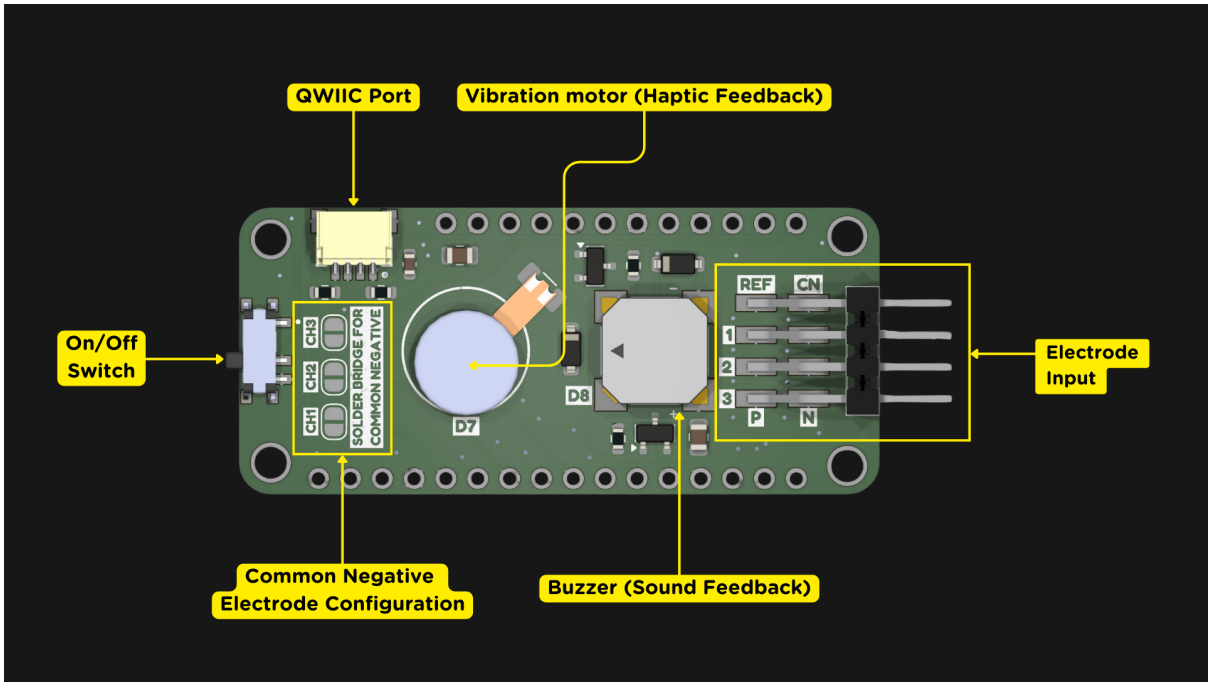



Fig. 16: VibZ Playmate

# 2 Skin Preparation Guide

## 2.1 Why skin preparation is important?

Proper skin preparation is crucial before recording any biopotential signal be it Electrocardiography (ECG), Electromyography (EMG), Electroencephalography (EEG), or Electrooculography (EOG).

- `Clean skin surface:` Removes dead skin cells, oils, & other substances that increases skin impedance.
- `Improve impedance:` Improves the conduction of electrical signals from the body to the recording equipment and minimizes impedance.
- `Electrode-skin contact:` Ensures optimal contact between the electrodes and the skin surface.
- `Signal quality:` Enhances the overall quality of recorded signals, providing clear & reliable data for analysis & improves the ability to capture subtle variations in biopotential signals.
- `Consistency in recordings:` Reduces variability in signal quality, making it easier to make any Human-Computer Interface (HCI), Brain-Computer Interface (BCI) project or a real-world application.
- `Long term adhesion:` Facilitates long-term adhesion & stable placement of electrodes to the skin during extended signal monitoring.

## 2.2 Kit Contents

| | |
|---|---|
| Nuprep gel | Mildly abrasive, highly conductive gel that should be applied before placing the electrodes on the skin to improve signal quality & enhances the performance of monitoring electrodes. |
| Electrode Gel | Highly conductive gel that acts as a coupling agent between dry electrodes and the skin to aid the transmission of biopotential signals like ECG, EMG, EOG, or EEG. |
| Ten20 paste | Contains the right balance of adhesiveness and conductivity, enabling the dry electrodes to remain in place while allowing the transmittance of biopotential signals. |
| Alcohol Swabs/Wet wipes | Soft & non-woven pads that helps in cleaning the skin surface and does not leave any residue. |
| Cotton Swabs | Useful while applying nuprep gel or ten20 paste. |

## 2.3 Steps to follow

You can follow the steps given below to do the skin preparation properly:

### 2.3.1 Step 1: Identify the targeted area

Identify the target area where the gel electrodes or BioAmp Bands will be placed for recording the biopotential signals.

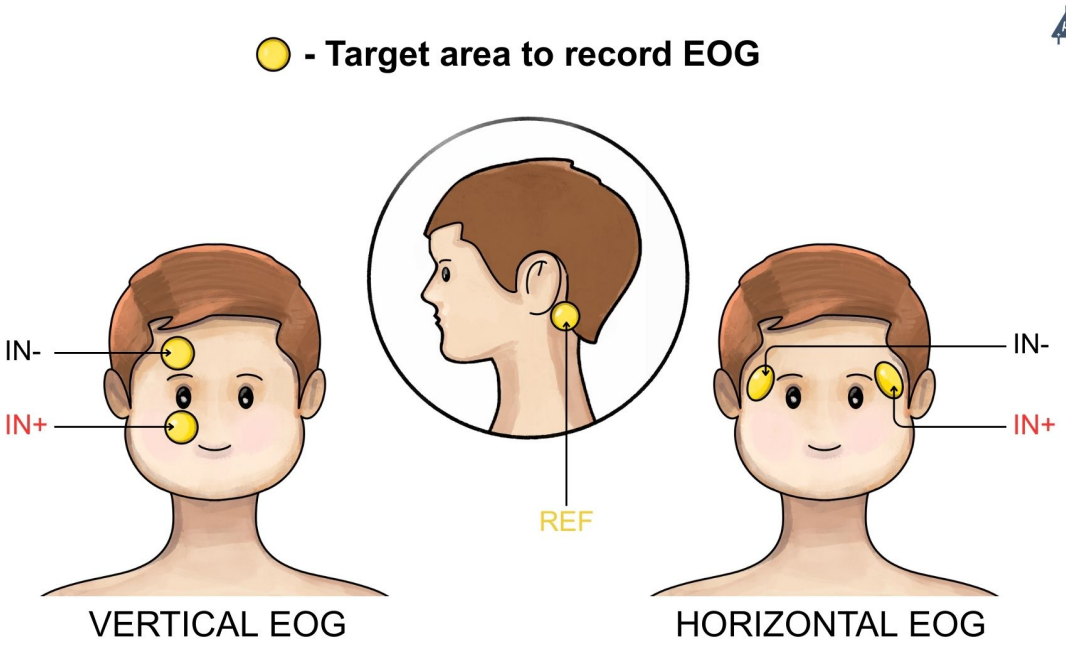



Fig. 1: Target area to record EOG

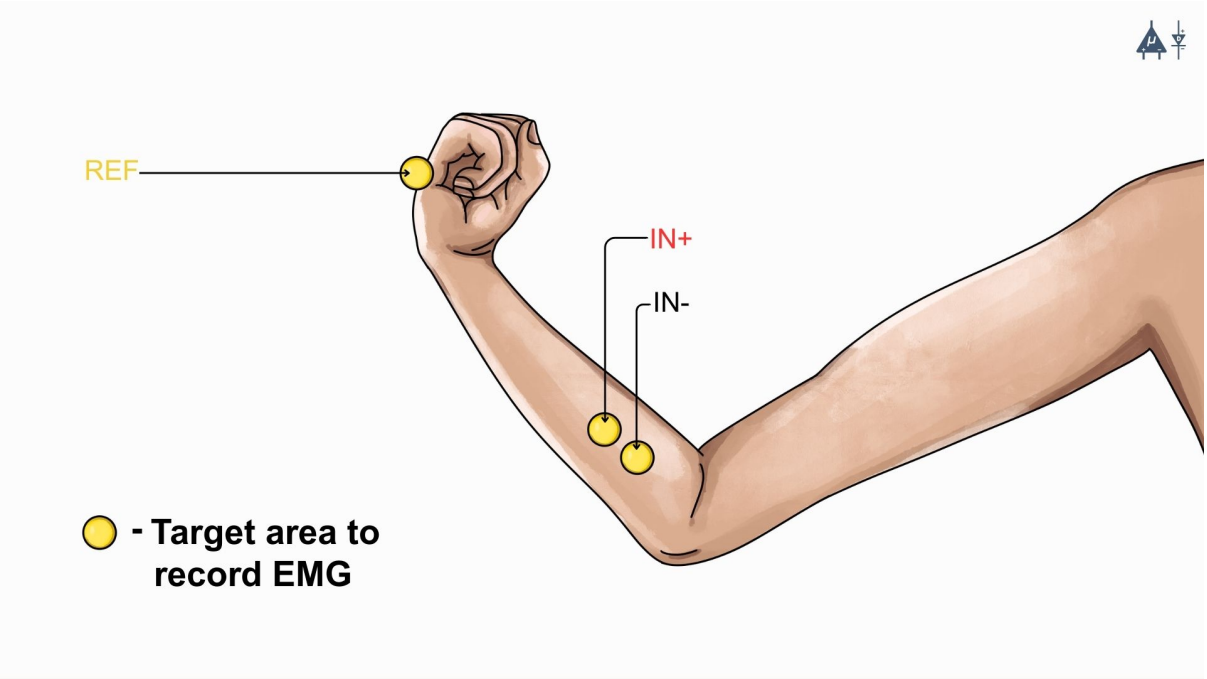



Fig. 2: Target area to record EMG

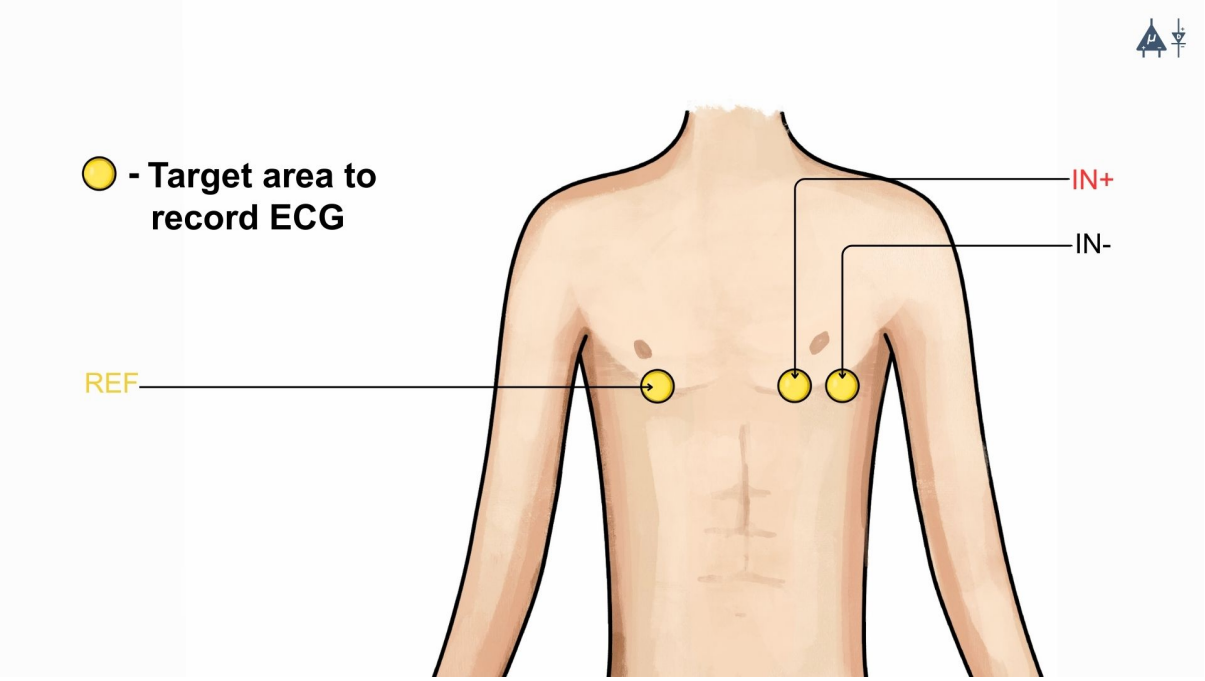



Fig. 3: Target area to record ECG

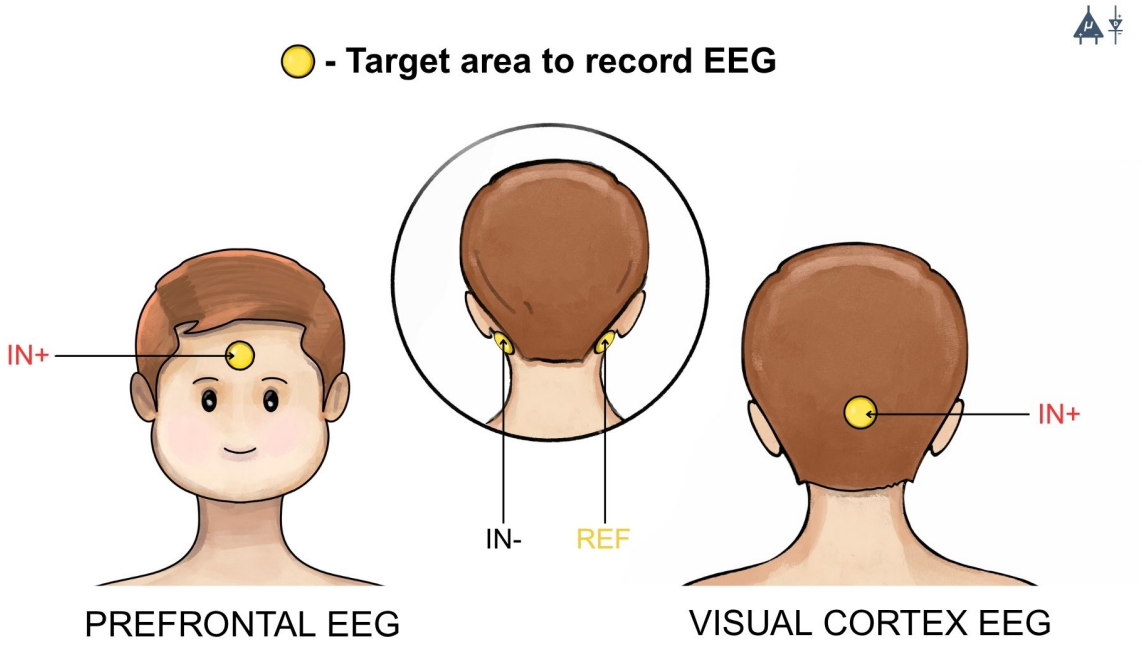



Fig. 4: Target area to record EEG

### 2.3.2 Step 2: Apply NuPrep gel

Take a small amount of NuPrep gel using a cotton swab and apply it on your targeted area.

### 2.3.3 Step 3: Clean the skin surface

Use gentle, circular motions to rub the gel on the skin surface. This removes all the dead skin cells & improves conductivity.

**Warning**

Do not rub the gel for too long as it has abrasive properties and may cause skin redness and irritation.

### 2.3.4 Step 4: Wipe off the gel

Wipe away excess gel with alcohol swabs or wet wipes.

**Warning**

- Using alcohol swabs can dry out the skin, so don't use them if your skin is already dry.
- Close your eyes while using the alcohol swabs for EOG recording else it may cause eye redness & irritation.

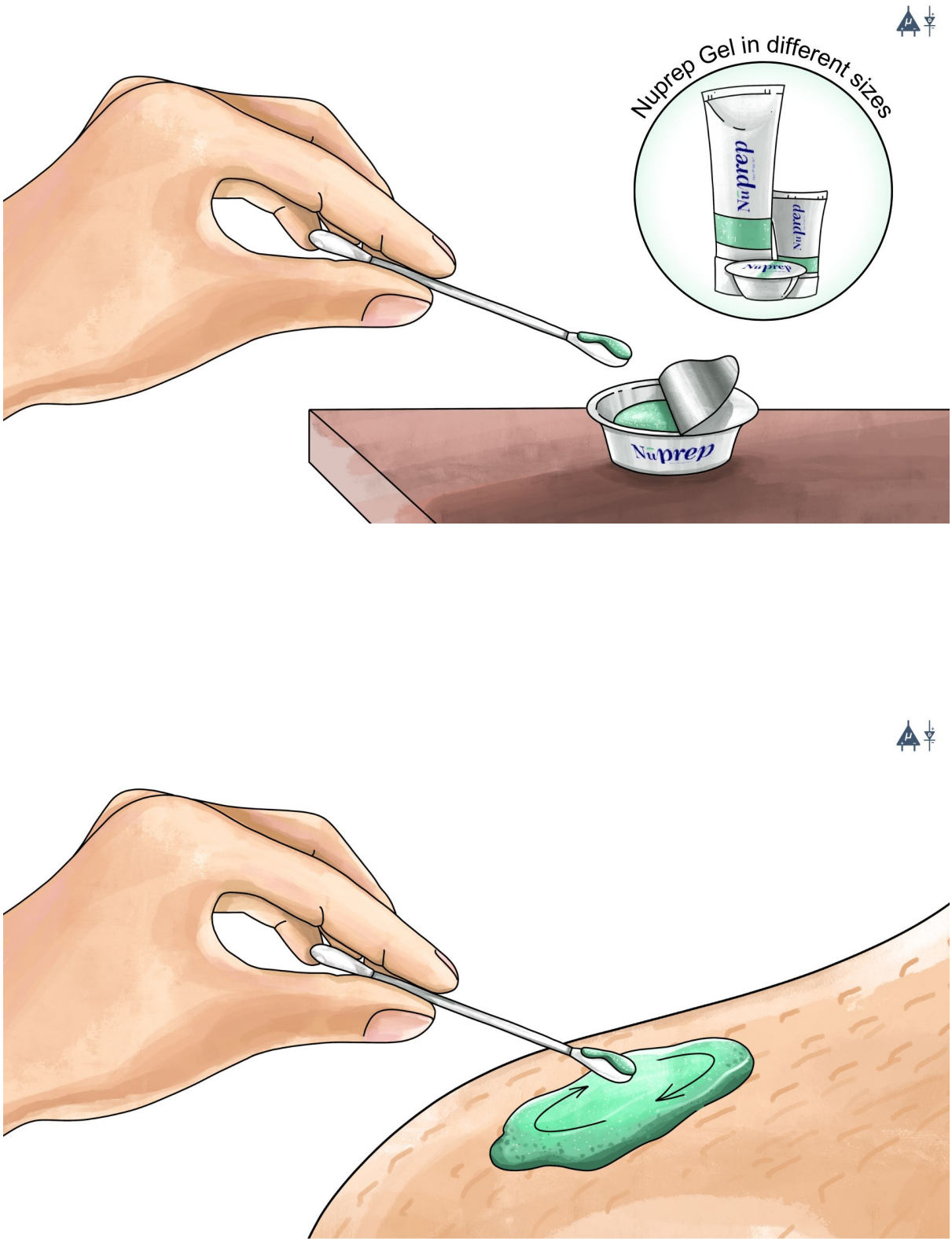



Fig. 5: Rub the gel gently using the cotton swab

Fig. 6: Wipe away excess gel

### 2.3.5 Step 5: Measuring the signals

Now you can either use gel electrodes or BioAmp bands for the signal recording.

#### Using gel electrodes

Connect the BioAmp cable to gel electrodes, peel the plastic backing from electrodes and place the IN+, IN-, REF cables according to your specific biopotential recording.

**Note**

While placing the gel electrodes on the skin, make sure to place the non-sticky tab of the electrode in the direction opposite to your hair growth. This allows you to remove the electrodes easily without pulling off much body hair.

#### Using BioAmp bands

Connect the BioAmp cable to your BioAmp band. Now apply a small amount of `electrode gel` or `Ten20 conductive paste` on the dry electrodes between the skin and metallic part of BioAmp cable. This improves the signal conductivity, enhancing overall signal quality.

**Note**

The above graphics demonstrates the use of electrode gel/Ten20 paste with Muscle BioAmp Band. Similarly you can use Brain BioAmp Band and Heart BioAmp Band. Refer to *Using BioAmp Bands* guide to assemble and use all the BioAmp Bands correctly.

Now you are all set! Make all the connections correctly and start recording your biopotential signals.

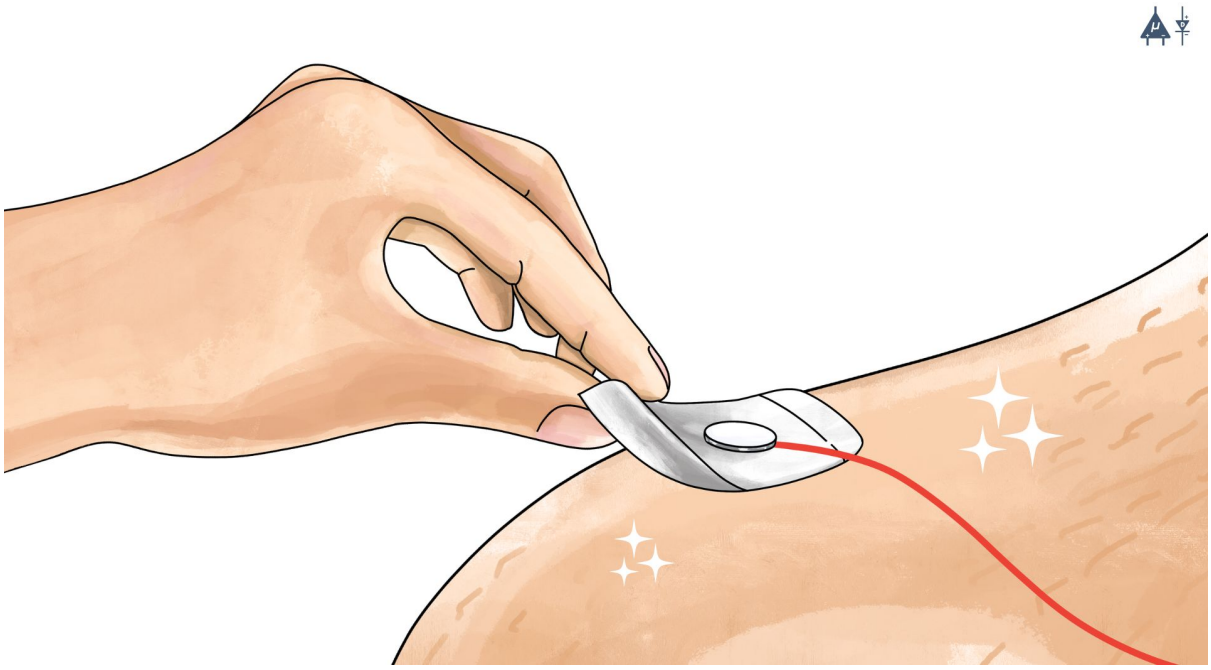


Fig. 7: Placing gel electrodes on skin surface

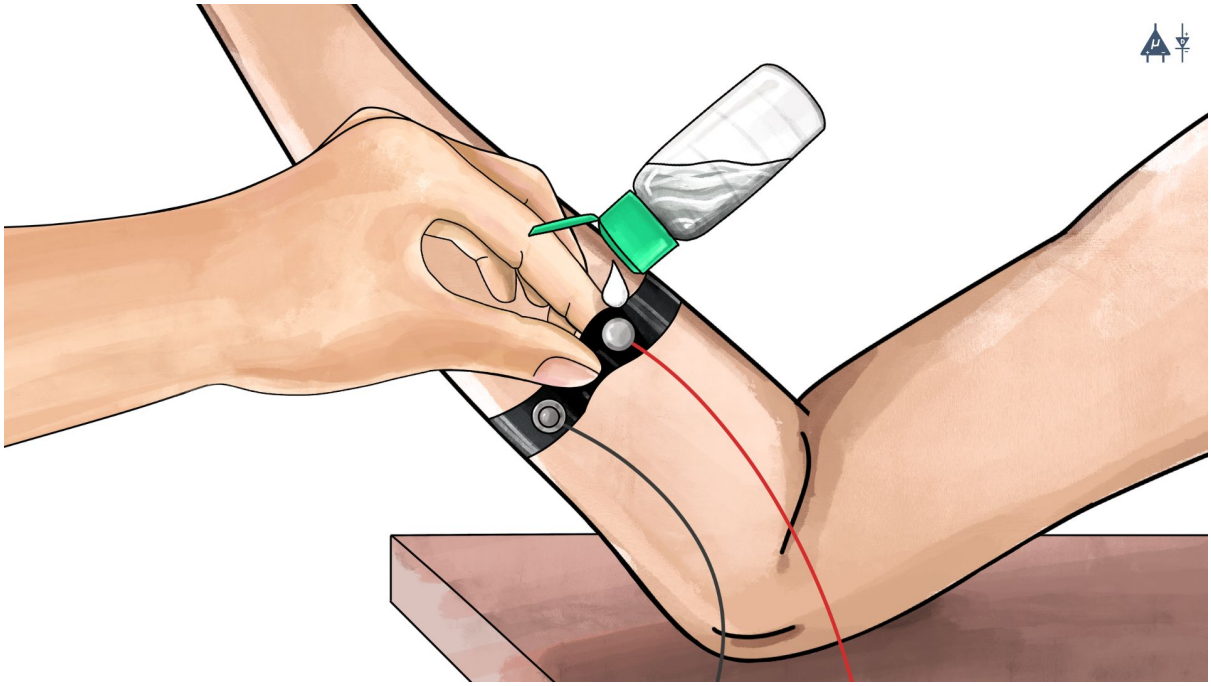


Fig. 8: Method 1: Using Electrode gel

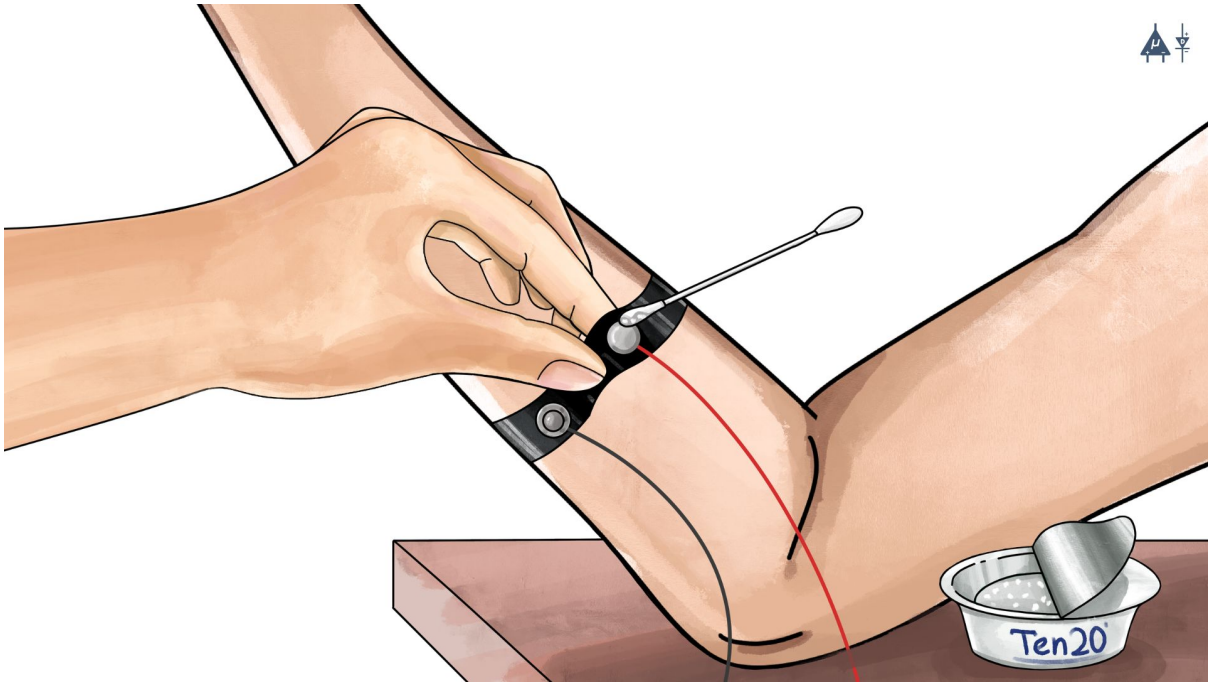



Fig. 9: Method 2: Using Ten20 paste

**Warning**

NuPrep gel, Ten20 paste and the alcohol swabs shouldn't be used if you have a history of skin allergies to lotions and cosmetics.

# 3 Using BioAmp Bands

## 3.1 Overview

BioAmp Bands are dry electrode-based stretchable bands that allows you to record biopotential signals from your body be it from brain (EEG), muscles (EMG) or heart (ECG). These bands can only be used with our BioAmp Hardware by making the connections using BioAmp Cable.

## 3.2 Why use BioAmp Bands?

Usually, people use gel electrodes to record biopotential signals from the skin surface. But, it has its own disadvantages. So we came up with these BioAmp Bands using which users can enjoy a more comfortable, cost-effective, and hassle-free experience while recording biopotential signals.

- Comfort: BioAmp Bands are generally more comfortable to wear than gel electrodes, especially for long-term recordings. They conform to the body's shape and avoid the sticky, sometimes irritating sensation of gel electrodes.
- Reusability: Unlike gel electrodes, which are often single-use and need to be replaced frequently, BioAmp Bands can be reused multiple times. This makes them more cost-effective and environmentally friendly.
- Ease of Use: These bands are easy to wear and adjust, reducing the hassle of setup and ensuring consistent placement.
- Hygiene: They can be easily cleaned and sanitized between uses, reducing the risk of skin irritation and infections. Gel electrodes, on the other hand, can leave residue on the skin surface.
- Performance: The bands can provide stable and reliable signal recordings depending on your environment conditions. For hot/humid conditions, the bands usually perform better while recording the signals. But if the weather is cold causing dry skin, then it is recommended to prepare the skin properly and apply electrode gel between the metallic part of cable and skin surface. If you feel that the skin impedence is increasing, then reapply electrode gel frequently. The other option is to use gel electrodes after preparing the skin properly.

## 3.3 Types of BioAmp Bands

There are 3 types of BioAmp Bands and all these bands offer targeted and efficient solutions for recording biopotential signals from the muscles, heart, and brain, making them versatile tools for a wide range of HCI/BCI applications.

### 3.3.1 1. Muscle BioAmp Band

Muscle BioAmp Band (EMG Band) is a stretchable band that can be connected to any of our Muscle BioAmp Hardware or any EXG sensor using a BioAmp Cable. It allows you to record your muscle signals hassle-free.

| | |
|---|---|
| Length | 13 inches |
| Stretchability | 2X (Upto 26 inches) |
| Usability | Reusable as it comes with washable fabric |
| Interface | Snap electrodes |
| Compatible Hardware | Muscle BioAmp Hardware or any EXG sensor |
| BioPotentials | EMG |
| No. of channels | 1 |
| Wearable | Yes |

### 3.3.2 2. Heart BioAmp Band

Heart BioAmp Band (ECG Band) is a stretchable band that can be connected to any of our Heart BioAmp Hardware or any EXG sensor using BioAmp Cable. It allows you to record your ECG signals hassle-free.

| | |
|---|---|
| Length | 37 inches |
| Stretchability | 2X (Upto 74 inches) |
| Usability | Reusable as it comes with washable fabric |
| Interface | Snap electrodes |
| Compatible Hardware | Heart BioAmp Hardware or any EXG sensor |
| BioPotentials | ECG |
| No. of channels | 1 |
| Wearable | Yes |

### 3.3.3 3. Brain BioAmp Band

Brain BioAmp Band (EEG Band) is a stretchable band that can be connected to any of our Brain BioAmp Hardware or any EXG sensor using BioAmp Cable to record signals from the brain hassle-free.

| | |
|---|---|
| Length | 15.5 inches |
| Stretchability | 2X (Upto 31 inches) |
| Usability | Reusable as it comes with washable fabric |
| Interface | Snap electrodes |
| Compatible Hardware | Brain BioAmp Hardware or any EXG sensor |
| BioPotentials | EEG |
| No. of channels | 2 or 6 |
| Wearable | Yes |

You can get either a 2-channel or a 6-channel Brain BioAmp Band according to your project or research requirements:

#### 2-Channel Brain BioAmp Band

It can be used to record EEG signals up to 2 channels either from the visual cortex (back of your head) or the prefrontal cortex part of brain.

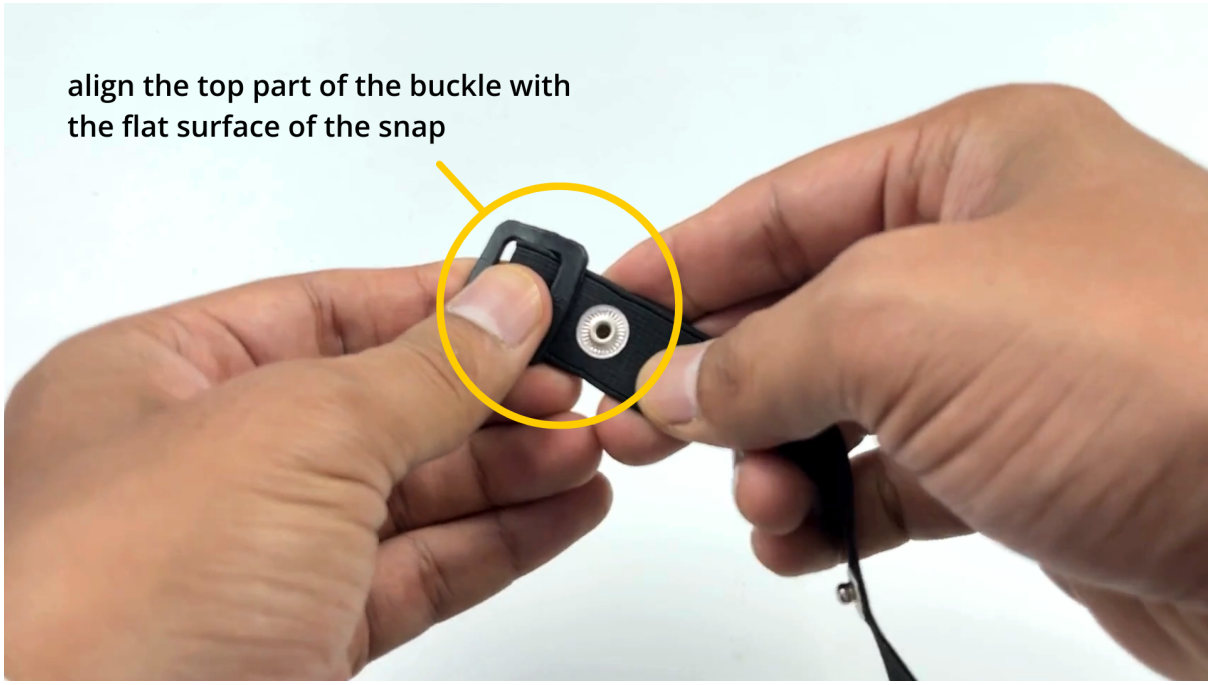
#### 6-Channel Brain BioAmp Band

It can be used to record EEG signals up to 2 channels either from the visual cortex (back of your head) or the prefrontal cortex part of brain.

## 3.4 Using Muscle BioAmp Band

### 3.4.1 Assembly

1. Take your Muscle BioAmp Band, hold the side of the band that has buckle on it and align the top part of the buckle with the flat surface of the snap.
1. Take the other end of the band and insert it in the buckle.
3. Your band is now ready to use. You can also adjust the size of the band according to your targeted muscle.


align the top part of the buckle with
the flat surface of the snap

Take other end of the band and
insert it in the buckle

Insert this end of the band into the buckle
as shown and pull it to finish the assembly

adjust the size of the band using this
loose end of the band

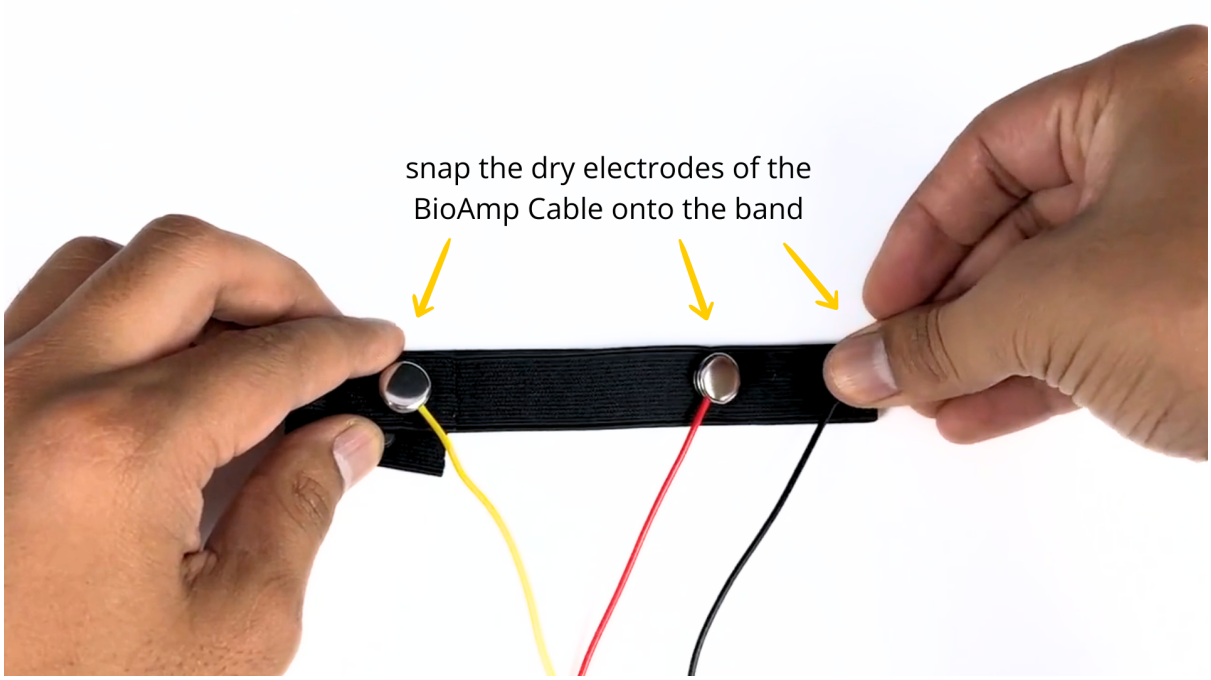
### 3.4.2 Skin Preparation

Apply Nuprep Skin Preparation Gel on the skin surface where dry electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step *Skin Preparation Guide*.

### 3.4.3 Measure EMG

1. Flip the band and snap the dry electrodes of the BioAmp Cable on it as shown below.




1. Flip the band again and wear it on your arm in such a way that IN+ and IN- are placed on the arm near the ulnar nerve and REF (reference) on the far side of the band.

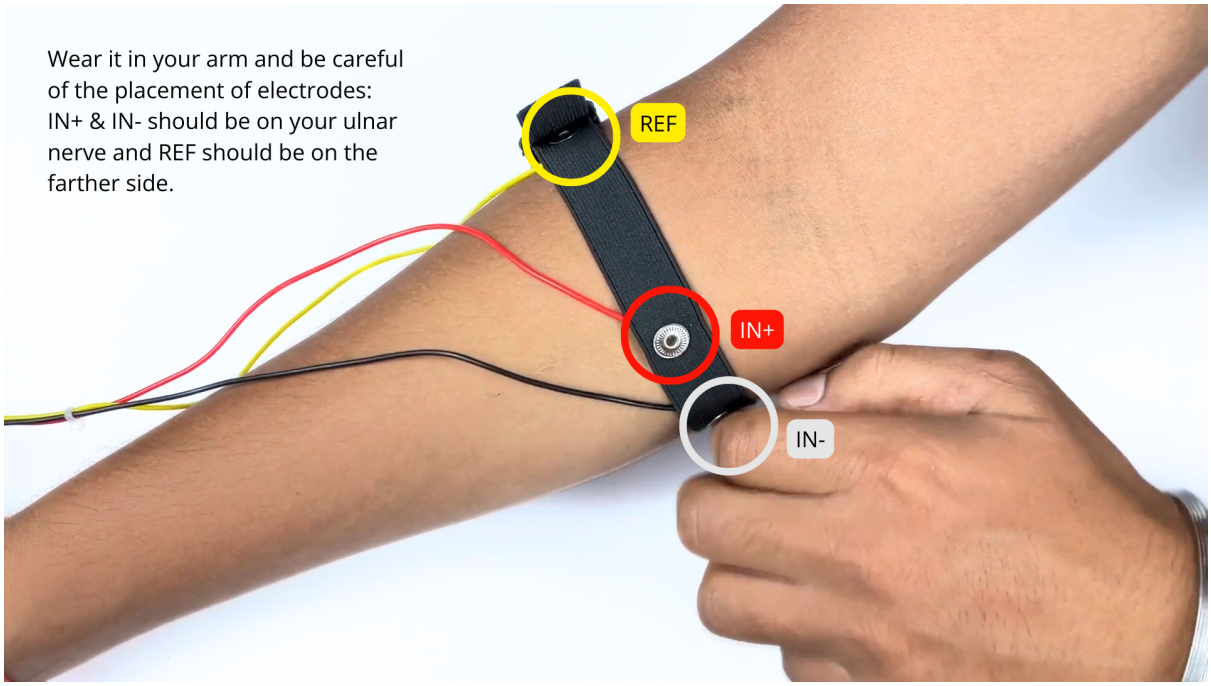
**Note**

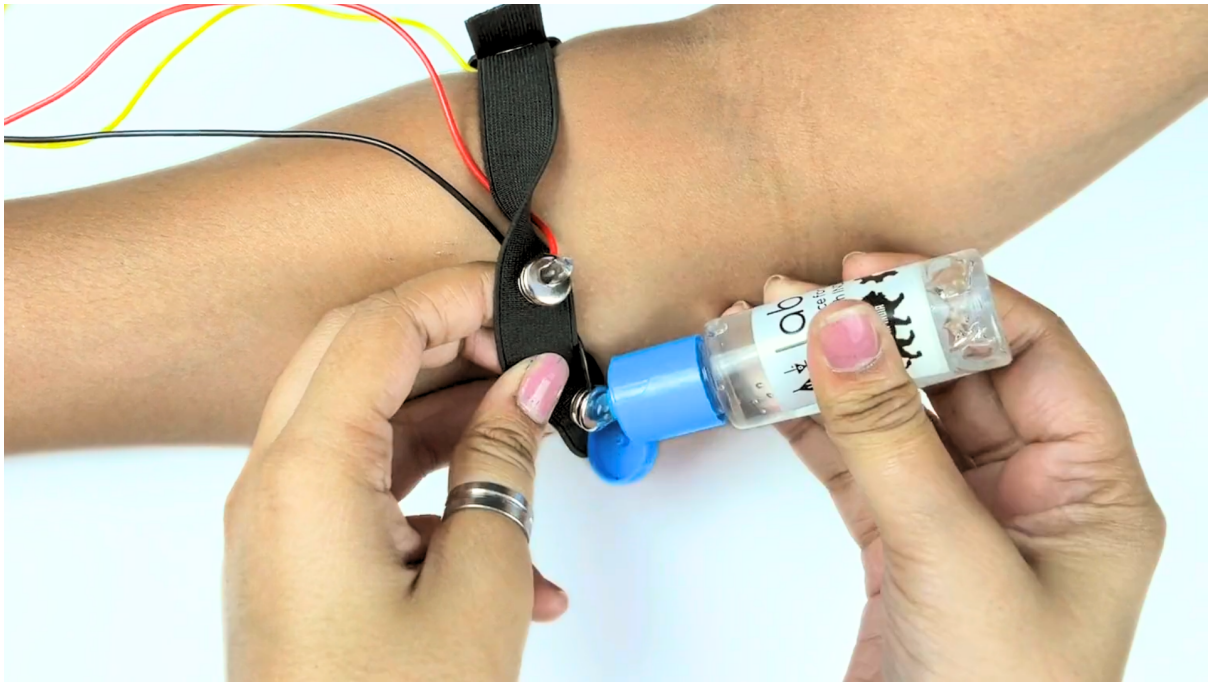
Make sure the dry electrodes (shiny parts of the BioAmp Cable) are in direct contact with the skin.

3. Now put a small amount of electrode gel or Ten20 paste between the skin and dry electrodes to get the best signal acquisition.

**Note**

- After using the band, don't leave the gel residue on the dry electrodes longer than an hour as it may corrode them over a period of time.
- Wash the band with liquid soap and rinse it properly after every use. Use it again only when it is completely dry.


Wear it in your arm and be careful of the placement of electrodes: IN+ & IN- should be on your ulnar nerve and REF should be on the farther side.
REF
IN+
IN-



## 3.5 Using Heart BioAmp Band

### 3.5.1 Skin Preparation

Apply Nuprep Skin Preparation Gel on your chest where dry electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step *Skin Preparation Guide*.

### 3.5.2 Assembly

1. Take your Heart BioAmp Band and wrap the band around your chest in such a way that the pointy part of the snap touches your chest and the flat part is on the outer side.

2. Now insert the loose end of the band into the buckle and tighten it by pulling the strap.
3. Your band is now ready to use. You can also adjust the size of the band according to your chest size.

### 3.5.3 Measure ECG

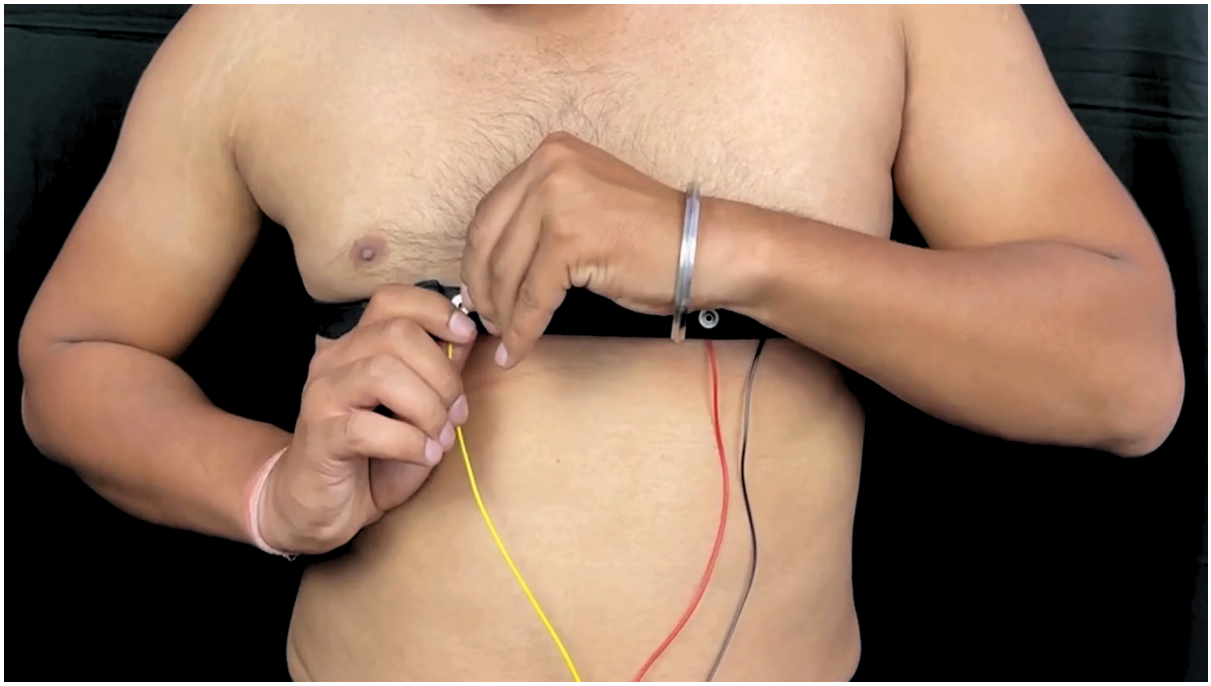
1. Snap the IN- cable on the left most side of the band, IN+ cable in the middle, and REF cable on the right side as shown below.

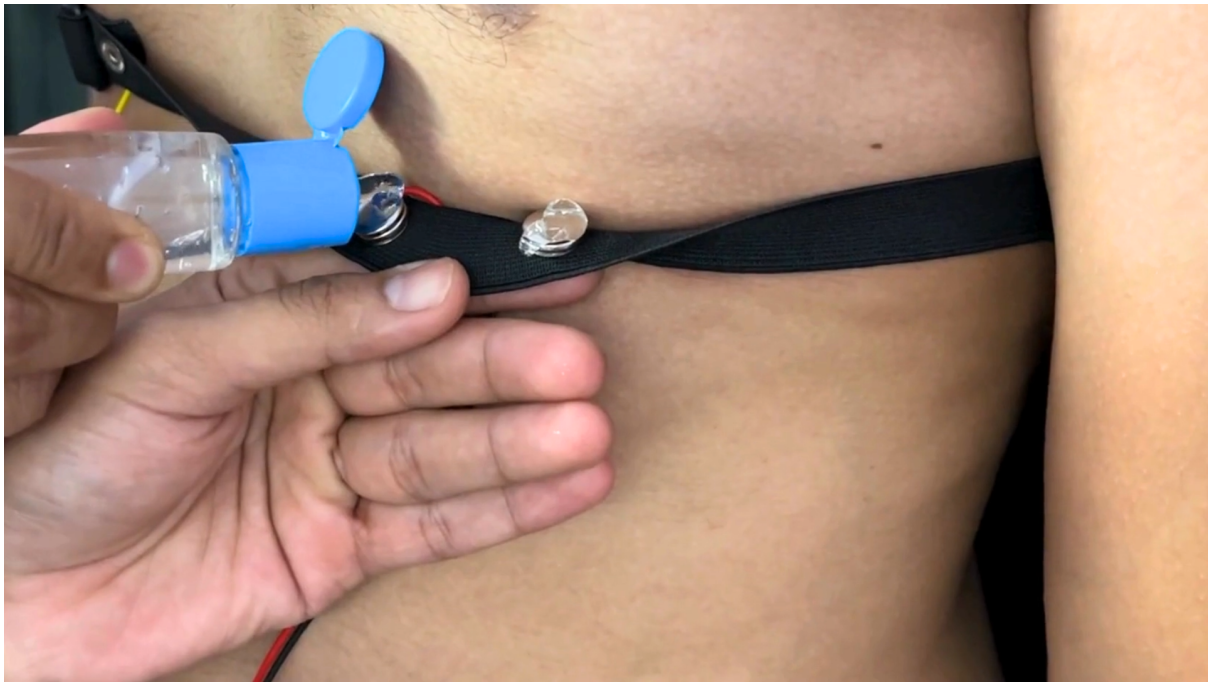
**Note**

Make sure the dry electrodes (shiny parts of the BioAmp Cable) are in direct contact with the skin.

2. Now put a small amount of electrode gel or Ten20 paste between the skin and dry electrodes to get the best signal acquisition.

**Note**





- After using the band, don't leave the gel residue on the dry electrodes longer than an hour as it may corrode them over a period of time.
- Wash the band with liquid soap and rinse it properly after every use. Use it again only when it is completely dry.

## 3.6 Using Brain BioAmp Band

### 3.6.1 Assembly

You get the band in two parts - the longer part consists of buckles at both ends and the shorter one has loose ends on both sides.

1. Hold one end of the longer band and align the top part of the buckle with the flat surface of the snap.
2. Now take the shorter band and insert it into the buckle of longer band.
3. Repeat step 1 and 2 for the other buckle on the longer band.
4. Your band is now ready to use. You can also adjust the size of the band according to your head size.

### 3.6.2 Skin Preparation

Apply Nuprep Skin Preparation Gel on your targeted area (visual cortex or prefrontal cortex) where dry electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step *Skin Preparation Guide*.

### 3.6.3 Measure 1-channel EEG

1. Flip the band, take your BioAmp Cable, and snap the REF cable on a gel electrode. Now snap the IN- and IN+ cable on:

- Fp1 and Fp2 positions for recording EEG from prefrontal cortex
- O1 and O2 positions for recording EEG from visual cortex

**Note**

The electrode positions mentioned above are according to International 10-20 sytem for recording EEG.

2. Flip the band again and wear it in a way so that the dry electrodes (shiny parts of the cable) are in contact with:

- skin surface on the forehead (if recording from prefrontal cortex)
- scalp surface on the back side of your head (if recording from visual cortex)

3. Peel of the plastic backing of the gel electrode and place it on the bony part behind your earlobe.

**Note**

While placing the gel electrodes on the skin, make sure to place the non-sticky tab of the electrode in the direction opposite to your hair growth. This allows you to remove the electrodes easily without pulling off much body hair.

4. Now put a small amount of electrode gel or Ten20 paste between the skin/scalp and dry electrodes to get the best signal acquisition.